ECON 131 - INTERNATIONAL FINANCIAL MARKETS

Pavitra Kanagaraj

10/12/2026

KEY TAKEAWAYS: LECTURES 4 - 5

LECTURE 4: LAW OF ONE PRICE & ABSOLUTE PPP

- **Law of One Price (LOOP)**: A good sells for the identical currency-denominated price everywhere. Violations: tariffs/quotas, transaction costs, noncompetitive markets, and sticky prices.
- **Internal vs. External Purchasing Power**:
  - Internal: Goods/services bought with \$1 domestically ($1/P_{\$}$).
  - External: Goods/services bought abroad after conversion ($1/[S \cdot P_{\text{for}}]$).
- **Absolute Purchasing Power Parity (PPP)**: Spot rate equalizes internal and external purchasing power:

$$S^{\text{PPP}}(\$/\text{FC}) = \frac{P_{\$}}{P_{\text{FC}}}$$

- **Big Mac Index**: If $S_{\text{market}} > S^{\text{PPP}}$, foreign currency is undervalued (or USD is overvalued).

LECTURE 4: RELATIVE PPP & REAL EXCHANGE RATES

- **Relative PPP**: Exchange rate percentage change offsets inflation differentials:

$$1 + s = \frac{1 + \pi_{\$}}{1 + \pi_{FC}} \implies s \approx \pi_{\$} - \pi_{FC}$$

- Higher domestic inflation causes the domestic currency to depreciate.
- **Real Exchange Rate (*RS*)**: Purchasing power ratio across countries:

$$RS(\$/FC) = \frac{S(\$/FC) \times P_{FC}}{P_{\$}}$$

- **Implication**:
  - If Absolute PPP holds, $RS = 1$.
  - A rise in *RS* represents a real depreciation of the domestic currency (foreign goods become relatively more expensive).

LECTURE 5: BALANCE OF PAYMENTS (BOP)

- **Balance of Payments (BOP)**: Accounting of all economic transactions between domestic residents and the rest of the world.
- **Double-entry accounting**: Every trade generates equal entries:
  - Credit (+): Inflows (Exports, foreign purchases of domestic assets).
  - Debit (–): Outflows (Imports, domestic purchases of foreign assets).
- **Current Account (CA)**: Trade balance (goods & services), net investment income, and unilateral transfers.
- **Capital / Financial Account (KA)**: Net purchases of foreign and domestic assets, including official reserve transactions.
- **Fundamental BOP Identity**:

$$\text{Current Account (CA)} + \text{Capital Account (KA)} = 0$$

LECTURE 5: NIPA LINKAGES & EXCHANGE RATE REGIMES

- **National Income Linkages** ($GNI = GDP + NFI$):

$$CA = GNI - (C + I + G) = S - I$$

  $CA < 0$ means domestic $I > S$, financed by foreign borrowing.
- **Exchange Rate Systems**:
  - **Floating**: Determined strictly by supply and demand (e.g., US, Japan, EU).
  - **Managed Floating**: Central Bank intervenes without explicit peg (e.g., Brazil, Mexico).
  - **Fixed / Pegged**: Pegged to single currency or basket (e.g., China).
  - **No Separate Legal Tender**: Dollarization (e.g., Ecuador, Panama).
- **FX Interventions**: Sterilized (offsetting domestic open market operations to neutralize monetary base) vs. Non-sterilized.

PS2 GUIDED PRACTICE

PROBLEM SET 2: PROMPT 1

- **Setup**:
  - As an importer of grain into Japan from the United States, you have agreed to pay **$326,434** in **120 days** after you receive your grain.
  - Spot rate: ¥103.89/$.
  - 120-day forward rate: ¥105.33/$.
  - 120-day USD interest rate: 3.56% p.a.
  - 120-day JPY interest rate: 2.27% p.a.

- **Tasks to Complete in Your Homework**:
  (a) Explain two ways to hedge the transaction foreign exchange risk.
  (b) Calculate the costs of the two alternatives in part (a) and decide which one is superior with a lower cost.

PROBLEM 1: PARALLEL EXAMPLE (HEDGING AN ACCOUNT PAYABLE)

- **Parallel Scenario**:
  - An Italian machinery importer owes **$250,000** in **90 days**.
  - Spot rate $S = \$1.10/€$.
  - 90-day forward rate $F = \$1.12/€$.
  - 90-day USD rate $i_{\$} = 4.0\%$ p.a., 90-day EUR rate $i_{€} = 2.0\%$ p.a.
- **(a) Two Hedging Alternatives**:
  1. **Forward Market Hedge**: Buy \$250,000 forward at $F = \$1.12/€$ today for delivery at $T = 90$.
  2. **Money Market Hedge**: Borrow euros today, convert to USD at spot, and deposit in a 90-day USD account to mature to \$250,000.

PROBLEM 1: PARALLEL EXAMPLE (COST COMPARISON)

- **De-annualize 90-day interest rates**:

$$i_{\$,90d} = 4.0\% \times \frac{90}{360} = 1.0\%, \quad i_{€,90d} = 2.0\% \times \frac{90}{360} = 0.5\%$$

- **Alternative 1 (Forward Hedge Cost at $T = 90$)**:

$$\text{Cost} = \frac{\$250,000}{\$1.12/€} = €223,214.29$$

- **Alternative 2 (Money Market Hedge Cost at $T = 90$)**:
  1. PV of USD needed today $= \frac{\$250,000}{1+0.010} = \$247,524.75$
  2. Euros borrowed at spot $= \frac{\$247,524.75}{\$1.10/€} = €225,022.50$
  3. Future value of euro loan $= €225,022.50 \times (1 + 0.005) =$ $€226,147.61$
- **Decision**: Forward Market Hedge is superior (saves €2,933.32).

PROBLEM SET 2: PROMPT 2

- **Setup**:
  - Price level in Canada: $P_{\text{Can}} = \text{CAD } 13,800$
  - Price level in France: $P_{\text{Fra}} = \text{EUR } 12,500$
  - Spot exchange rate: $S = \text{CAD } 1.20/\text{EUR}$

- **Questions to Solve**:
  1. What is the internal purchasing power of 10,000 Canadian dollars?
  2. What is the internal purchasing power of 10,000 euros in France?
  3. What is the implied exchange rate of CAD/EUR satisfying absolute PPP?
  4. Is the euro overvalued or undervalued relative to the Canadian dollar?
  5. What amount of appreciation or depreciation of the euro would be required to return the actual exchange rate to its PPP value?

PROBLEM 2: PARALLEL EXAMPLE (BOOK CH 8, PROBLEM 3)

- **Given Data (Book Problem 3)**:
  - Price level in Canada: $P_{\text{Can}} = \text{CAD } 16,600$
  - Price level in France: $P_{\text{Fra}} = \text{EUR } 11,750$
  - Spot exchange rate: $S = \text{CAD } 1.35/\text{EUR}$

- **(a) Internal purchasing power of 1 CAD**:

$$\text{PP}_{\text{Can}} = \frac{1}{P_{\text{Can}}} = \frac{1}{\text{CAD } 16,600} = 0.00006024 \text{ baskets per CAD}$$

- **(b) Internal purchasing power of 1 EUR in France**:

$$\text{PP}_{\text{Fra}} = \frac{1}{P_{\text{Fra}}} = \frac{1}{\text{EUR } 11,750} = 0.00008511 \text{ baskets per EUR}$$

PROBLEM 2: PARALLEL EXAMPLE (PARTS C, D, AND E)

- **(c) Implied Absolute PPP Exchange Rate**:

$$S^{\text{PPP}}(\text{CAD/EUR}) = \frac{P_{\text{Can}}}{P_{\text{Fra}}} = \frac{\text{CAD } 16,600}{\text{EUR } 11,750} = \text{CAD } 1.4128/\text{EUR}$$

- **(d) Valuation of the Euro**:
  - Market rate: $S = \text{CAD } 1.35/\text{EUR}$
  - PPP benchmark: $S^{\text{PPP}} = \text{CAD } 1.4128/\text{EUR}$
  - Since $S < S^{\text{PPP}}$, the Euro is undervalued relative to the Canadian dollar (EUR buys fewer CAD than price levels dictate).
- **(e) Required % Change to Restore PPP**:

$$\%\Delta = \frac{S^{\text{PPP}} - S}{S} = \frac{1.4128 - 1.35}{1.35} = \frac{+0.0628}{1.35} = +4.65\%$$

- The euro must appreciate by 4.65% against CAD to reach parity.

SUBMISSION REMINDER FOR STUDENTS

- **Due Date**: In class on Tuesday.
- **Format**: Handwritten work only.
- **Late policy**: No late submissions or email submissions accepted.
- **Requirement**: Write out all formulas, unit conversions, and operational steps explicitly as shown in these slides.